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B.Tech. Degree VII Semester Examination November 2017

EB/EC 1705 (E4) MECHATRONICS

(2012 Scheme)

Time: 3 Hours

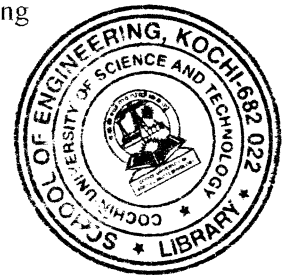
Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define mechatronics. Discuss the key elements of mechatronics system.
- (b) Explain the fluid system building blocks.
- (c) A hydraulic cylinder with piston diameter of 150 mm is used to move a work piece in a manufacturing operation through a distance of 250 mm in 15 sec. The force required to move the work piece is 50 kN. Find the working pressure and hydraulic liquid flow rate required to do the operation.
- (d) Explain the working of cams and its uses.
- (e) Distinguish repeatability and accuracy of robotic manipulator.
- (f) Explain the methods of improving machine accuracy of NC machines.
- (g) Describe MEMS actuator and sensor with the help of block diagram.
- (h) Describe about the fibre optic devices in mechatronics.



PART B

(4 × 15 = 60)

- II. What do you mean by system modeling? Develop a mathematical model for armature controlled DC motor. (15)
- OR**
- III. (a) Develop a mathematical model for hydraulic linear actuator. (10)
- (b) What is the overall transfer function of a closed loop system having a forward path having 2 series elements with $10 & \frac{1}{s+2}$ and a negative feedback path transfer function of $2/s$? (5)
- IV. Sketch and explain the working of following feedback devices (15)
 - (i) Encoders (ii) Resolvers (iii) Inductosyn (iv) Tachometers.
- OR**
- V. (a) Explain the working of hydraulic pumps with suitable diagram. (9)
- (b) Sketch and explain the working of lead screw, rack and pinion mechanism. (6)
- VI. (a) Explain the different classification of NC systems. (7)
- (b) Discuss the different aspects to be considered in the design of NC machine tools. (8)
- OR**
- VII. (a) Explain the different classification of robotic manipulator with diagram. (10)
- (b) Compare the merits and demerits of NC and CNC machines. (5)
- VIII. (a) Explain the structure of PLC. (10)
- (b) What are the different applications of MEMS? (5)
- OR**
- IX. Explain the effect of adding On/Off, P, PI, PD and PID controllers in the design of servo motor. (15)